1 Question (15 points)

Find the square root of an integer N using divide and conquer technique which run in $\mathcal{O}(\log N)$ time. You are not allowed to use sqrt function.

2 Question (15 points)

Let A $= \{a_1, a_2, \ldots, a_n\}$ and B $= \{b_1, b_2, \ldots, b_n\}$ be two sets of numbers. Consider the problem of finding their unions, i.e., the set C of all the numbers that are in A or B. Solve the problem in $\mathcal{O}(n \log n)$ time.

3 Question (15 points)

There is a row of n coins whose values are some positive integers $c_1, c_2, \ldots, c_n$, not necessarily distinct. The goal is to pick up the maximum amount of money subject to the constraint that no two coins adjacent in the initial row can be picked up. Solve the instance 5, 3, 2, 8, 6, 4, 7, 6, 1, 5 of this coin-row problem.

4 Question (15 points)

You are given an n by n board. Write a dynamic programing algorithm to find the number of possible paths from (1,1) to (n, n). There are also holes in the board, where you can not pass. The positions of the holes are given by the matrix A, where A[i][j] is 1 if there is a hole in row i col j; otherwise A[i][j] is 0.